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TECHNOLOGICAL UNIVERSITY DUBLIN

CITY CAMPUS - GRANGEGORMAN

**Master of Science in Computer Science –
(Advanced Software Development - Full-Time)**

**Master of Science in Computer Science –
(Data Science - Part-Time)**

Year 1

SUPPLEMENTAL - EXAMINATIONS 2024/25

Data Mining

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Special Instructions/Handouts: No calculators or other devices should be used. There is no need.

- Q1. (a)** Describe the role of data preprocessing in data mining and list three techniques used to prepare data for analysis.

(10 marks)

Role of Data Preprocessing: Data preprocessing is essential in data mining as it improves the quality of data and enhances the efficiency of the mining process. It involves cleaning, transforming, and organizing data to make it suitable for analysis.

Techniques:

- **Data Cleaning:** Removing noise and correcting inconsistencies in the data (e.g., handling missing values, removing duplicates).
- **Normalization:** Scaling numerical data to a standard range (e.g., Min-Max scaling or Z-score normalization) to ensure uniformity across features.
- **Feature Selection:** Selecting relevant features to reduce dimensionality, which helps in improving model performance and reducing overfitting.

- (b)** Explain the concept of Supervised Learning and provide examples of tree-based algorithms commonly used in this approach, highlighting their key differences.

(10 marks)

Concept of Supervised Learning: Supervised learning is a type of machine learning where a model is trained on a labelled dataset, meaning that each training example includes input-output pairs. The model learns to map inputs to the correct outputs.

Examples of Tree-Based Algorithms:

1. **Decision Trees:** Simple, interpretable models that split data into subsets based on feature values, producing a tree-like structure.
2. **Random Forests:** An ensemble method that combines multiple decision trees to improve predictive accuracy and control overfitting.
3. **Gradient Boosting Trees:** Builds trees sequentially, where each new tree corrects errors made by the previous ones, often leading to better performance but at the cost of interpretability.

Key Differences:

- **Complexity:** Decision trees are simple and easy to interpret, while Random Forests and Gradient Boosting Trees are more complex and provide better accuracy.
- **Robustness:** Random Forests are less prone to overfitting compared to individual decision trees due to averaging multiple trees.

- (c)** Differentiate between overfitting and underfitting in the context of machine learning models. Provide examples of each.

(10 marks)

Overfitting: Overfitting occurs when a model learns the training data too well, capturing noise and outliers, which leads to poor generalization to unseen data.

- **Example:** A complex polynomial regression that fits every data point perfectly but performs poorly on test data.

Underfitting: Underfitting happens when a model is too simple to capture the underlying patterns in the data, leading to poor performance on both training and test sets.

- **Example:** A linear regression model applied to a nonlinear dataset, resulting in significant prediction errors.

Q2 (a) Discuss how the concept of entropy is used in decision trees to make splits.

(10 marks)

Entropy is a measure of impurity or disorder in a dataset. In decision trees, it quantifies the uncertainty in predicting the class label of a dataset. When building a decision tree, the algorithm aims to minimize entropy when splitting the data.

1. **Calculation:** The entropy of a dataset is calculated using the formula:

$$\text{Entropy}(S) = - \sum_{i=1}^C p_i \log_2(p_i)$$

where p_i is the proportion of class i in the set S and C is the number of classes.

2. **Splitting Criteria:** The algorithm evaluates potential splits based on how much they reduce entropy (Information Gain). A split with a higher Information Gain is preferred as it results in purer child nodes.

(b) Describe the concept of dimensionality reduction and explain two techniques used for this purpose. Outline the main differences between wrappers and filter methods.

(10 marks)

- **LIME (Local Interpretable Model-agnostic Explanations):** Generates locally faithful explanations by creating a simpler model around the prediction of interest, providing instance-level interpretations.
- **SHAP (SHapley Additive exPlanations):** Provides global and local interpretability by calculating the contribution of each feature to predictions based on game-theoretic Shapley values.

(c) Indicate if each statement is true or false and provide an explanation for your choice:

1. Unsupervised Learning does not require labeled data for training.
2. The use of data integration can lead to improved data quality.
3. Random Forests are less robust to outliers than Decision Trees.
4. Clustering algorithms can only be used on numerical data.

(10 marks)

1. **True.** Unsupervised learning analyses data without labels, identifying patterns and structures.
2. **True.** Data integration combines data from different sources, which can enhance quality by providing a more comprehensive view.

3. **False.** Random Forests are generally more robust to outliers because they aggregate multiple trees, which can mitigate the impact of outliers.
4. **False.** While many clustering algorithms work with numerical data, techniques like k-modes can handle categorical data.

Q3 Answer four out of the five next points:

- (a) Explain the concept of time series analysis and describe one method used to forecast future values. Provide a practical application example.

(10 marks)

- **Concept of Time Series Analysis:** Time series analysis involves statistical techniques to analyze time-ordered data points. It aims to understand underlying patterns, trends, and seasonal variations over time.
- Method for Forecasting:**
- **ARIMA (AutoRegressive Integrated Moving Average):** A widely used forecasting method that models time series data using autoregressive and moving average components along with differencing to achieve stationarity.
 - **Practical Application Example:** Forecasting stock prices or sales figures for businesses based on historical data, allowing companies to make informed decisions on inventory and investment.

- (b) Describe the differences between Local and Global explainability methods in the context of machine learning models. Give an example of each.

(10 marks)

- Local Explainability Methods:** Local explainability focuses on interpreting individual predictions made by a model, often using techniques to explain specific instances.
- **Example:** LIME (Local Interpretable Model-agnostic Explanations) provides explanations for individual predictions by approximating the model locally.
- Global Explainability Methods:** Global explainability aims to provide an overall understanding of the model's behavior and the relationships between features and predictions across the entire dataset.
- **Example:** Feature importance scores from a Random Forest model indicate which features are most influential for predictions on average.

- (c) Compare and contrast Bag of Words and word embeddings in Natural Language Processing (NLP). Discuss the strengths and weaknesses of each approach in a given NLP task.

(10 marks)

Bag of Words (BoW):

- **Definition:** Represents text as a set of words, disregarding grammar and order, using a vector where each dimension corresponds to a word's frequency.
- **Strengths:** Simple to implement and interpret; effective for document classification.
- **Weaknesses:** Loses semantic meaning and context; results in high-dimensional sparse vectors.

Word Embeddings:

- **Definition:** Represents words in dense vector spaces, capturing semantic relationships and contexts (e.g., Word2Vec, GloVe).
- **Strengths:** Encodes meaning and relationships between words, improving performance in tasks like sentiment analysis and translation.
- **Weaknesses:** Requires a large corpus for training; less interpretable than BoW.

- (d) Compare and contrast classification and regression algorithms. Provide examples of algorithms used in each category and discuss their typical applications.

(10 marks)

- Classification and regression algorithms are both types of supervised learning, but they differ in their output types and applications. Classification algorithms predict discrete labels, such as categories or classes. Examples include Logistic Regression, which is used for binary or multiclass classification, and Decision Trees, which segment data into subsets to make predictions. Classification is commonly used in applications like email spam detection, image recognition, and medical diagnosis.
- Regression algorithms, on the other hand, predict continuous numerical values. Linear Regression, which predicts a dependent variable based on a linear relationship with independent variables, and Decision Tree Regression, which segments data for predicting continuous outputs, are examples of regression algorithms. Typical applications include house price prediction, stock price forecasting, and temperature estimation, where predicting precise values is essential.

- (e) Describe the role of neural networks in data mining. Explain how a basic neural network is structured and provide an example of a problem it could solve.

(10 marks)

Role of Neural Networks: Neural networks are powerful machine learning models used for data mining that can capture complex patterns in large datasets. They consist of interconnected layers of nodes (neurons) that process inputs to produce outputs.

Basic Structure:

1. **Input Layer:** Receives input features.
2. **Hidden Layers:** One or more layers that perform computations and feature extraction.
3. **Output Layer:** Produces the final prediction.

Example Problem: A neural network can be used for image recognition tasks, such as class.

